VEERA NAGA KOUSHIK CHALADI

AI / GENERATIVE AI ENGINEER | Agentic AI | RAG | LLM Applications | Python | AWS
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PROFESSIONAL SUMMARY

AI/GenAI Engineer with 5+ years of software engineering experience building enterprise-grade AI systems across agentic workflows, RAG, scientific evidence reasoning, decision intelligence, and scalable backend services. Hands-on experience with Python, FastAPI, LangGraph, LangChain, AWS, LLM orchestration, retrieval optimization, guardrails, and observability. Experienced in architecture, production deployment, performance optimization, and engineering delivery.

TECHNICAL SKILLS

Generative & Agentic AI: LLMs, Agentic AI, Multi-Agent Systems, Plan-and-Execute, ReAct, Agentic RAG, Tool Calling, Replanning, Human-in-the-Loop, Guardrails, MCP
Frameworks: LangChain, LangGraph
RAG & Retrieval: Hybrid Search, Semantic Search, Embeddings, Chunking, Metadata Filtering, Query Rewriting, Reranking, Context Optimization, Retrieval Evaluation
LLM Fine-Tuning - Working Knowledge: Hugging Face, SFT, PEFT, LoRA, QLoRA, Quantization
Predictive AI: Temporal Feature Engineering, Risk Scoring, Forecasting, Explainability, What-if Analysis
Backend & Data: Python, FastAPI, AsyncIO, REST APIs, Microservices, System Design, PostgreSQL, pgvector, DynamoDB, Vector Databases, OAuth2, AWS SNS
Cloud & LLMOps: AWS, EC2, ECS, EKS, RDS, Docker, Kubernetes, CI/CD, MLflow, OpenTelemetry, CloudWatch, Grafana

PROFESSIONAL EXPERIENCE

AMGEN | Senior Associate Software Engineer | Jun 2025 - Present

Project: Scientific Results Reasoning Agent - Precision Medicine & Evidence Intelligence (Primary Project)

- Architected an **agentic scientific-reasoning system** that converts complex analysis outputs into evidence-grounded Precision Medicine insights and reviewer-ready briefs.
- Synthesized differential gene-expression, pathway-enrichment, immune-signature, correlation, and survival-style analyses across genes, pathways, immune signals, and clinical associations.
- Connected **PubMed**, public literature, internal knowledge, **vector retrieval**, and **MCP tools** to support traceable scientific interpretation.
- Cut **LLM context and token usage by 20%** through selective retrieval, evidence deduplication, reranking, and context optimization.
- Eliminated **10% of redundant LLM calls** by reusing structured intermediate findings across retrieval, synthesis, and biological reasoning.
- Lowered **end-to-end reasoning latency by 15%** through parallel evidence retrieval, selective tool invocation, and context reuse.
- Mapped analytical observations to biological mechanisms, treatment-response patterns, resistance hypotheses, biomarker candidates, and mechanism-of-action context.
- Produced source-linked briefs with key findings, supporting evidence, biological interpretation, MoA context, and follow-up questions for SME review.

Project: HRMAP AI - Enterprise Agentic AI & RAG Platform

- Lead architecture and backend delivery for an **enterprise agentic AI assistant** that generates employee reviews from Workday data, feedback, recognitions, conversation history, and uploaded documents.
- Own implementation planning, design/code reviews, technical decisions, and engineering guidance for the AI/backend workstream.
- Orchestrate **multi-step workflows** for intent handling, retrieval, generation, revision, document Q&A, and grounded responses.
- Trim LLM context through metadata-aware retrieval, reranking, selective context injection, and query optimization.
- Accelerate responses with **asynchronous execution**, **streaming**, selective tool invocation, guardrails, and validation that prevent unnecessary model calls.
- Productionized **Python/FastAPI services** with persistence, failure handling, deployment, and observability.
- Cut **AWS infrastructure spend by 29.4%** and **DynamoDB cost by an additional 10%** through resource and architecture optimization.

Project: TWFS Execution Readiness Command Center (Side AI Initiative)

- Co-developed a **predictive execution-intelligence platform** forecasting hiring, onboarding, proficiency, critical-mass, and capability go-live risks.
- Separated canonical data, temporal features, predictive analytics, decision logic, and LLM explanation into **auditable architecture layers**.
- Defined forecasts for hiring SLA breaches, onboarding dates, proficiency, critical-mass readiness, go-live risk, bottlenecks, and risk deterioration.
- Grounded explanations in measurable evidence, model contributions, business thresholds, reason codes, and recommended actions instead of free-form LLM reasoning.
- Constrained the agent layer to **controlled evidence packets** and analytics APIs for Q&A, executive summaries, and what-if analysis; helped productize the hackathon prototype.

PEGASYSTEMS | Aug 2021 - Jun 2025

Applications Engineer | Feb 2023 - Jun 2025

Project: Pega AutoPilot - Agentic AI & Advanced RAG Platform

- Engineered an AI-driven automation platform using Python, FastAPI, **Plan-and-Execute orchestration**, multi-agent workflows, and RAG.
- Created a **planner-executor workflow** that decomposed requests, selected tools/retrieval operations, coordinated execution, and replanned on failure.
- Applied **semantic/hybrid retrieval**, embeddings, metadata filtering, query rewriting, reranking, and targeted context construction to improve relevance while limiting context-window usage.
- Modularized tools and retrievers to separate orchestration from application capabilities and reduce repeated prompt/context construction.
- Raised **automation success rate by 20%** through orchestration, retrieval, and tool-execution enhancements.
- Cut **LLM API cost by 35%** through token optimization, shorter context payloads, streamlined function execution, and fewer model calls.

Associate Applications Engineer | Aug 2021 - Jan 2023

Project: Smart Claims Engine

- Developed **backend workflows** and **REST integrations** to support enterprise claims processing and downstream system connectivity.
- Implemented claims-processing **business logic**, **data validation**, and **exception handling** for reliable workflow execution.
- Implemented **near-real-time indexing** and optimized **enterprise search** to improve claims discovery and information retrieval.
- Improved service reliability through error handling, **structured logging**, **API testing**, and **performance tuning**.
- Built **data-processing components** and optimized **claims retrieval** and **reporting operations**.
- Supported **backend integration troubleshooting** and resolved workflow-level issues to improve consistency across claims-processing services.

CERTIFICATIONS & AWARDS

Certifications: Google TensorFlow Developer Certification (2023) | Pega Certified Senior System Architect (2021)

Awards: Pega 4x4 Award Winner | Two-time Spot Award Winner

EDUCATION

B.Tech - Information Technology | Velagapudi Ramakrishna Siddhartha Engineering College (VRSEC) | GPA: 8.45/10 | 2017 - 2021